Chronic Obstructive Pulmonary Disease (COPD)

السدة الرئوية المزمنة

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Outline

- Aetiology
- Pathogenesis and pathology
- Symptoms and Signs
- Clinical types
- Diagnosis
- Classification
- Co-morbidities
- Complications
- Differential Diagnosis
- Investigations
- Treatment

Definition

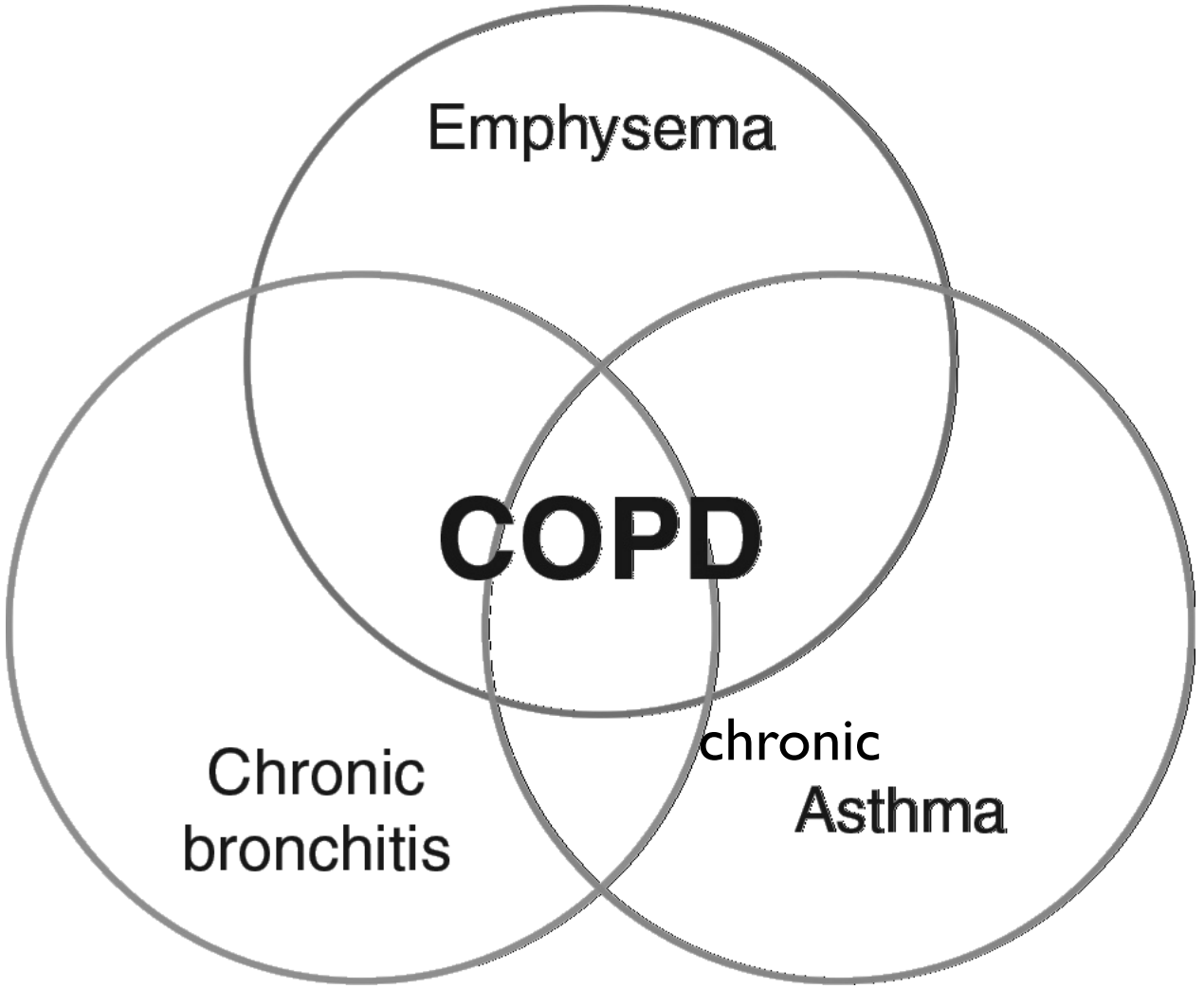
- COPD is a common chronic pulmonary disease with high morbidity and mortality. It is the third most common cause of death worldwide
- It is characterized by persistent respiratory symptoms (cough, sputum, dyspnea) and airflow limitation which is caused by a mixture of small airway obstruction and parenchymal destruction.

Aetiology (risk factors)

1. **Smoking** (80–90% of cases):
   - 10–15% of smokers develop COPD.
   - Smoking index (number of pack-years of smoking) correlates to reduction of FEV1.
   - Also, passive smokers are at increased risk to get COPD.

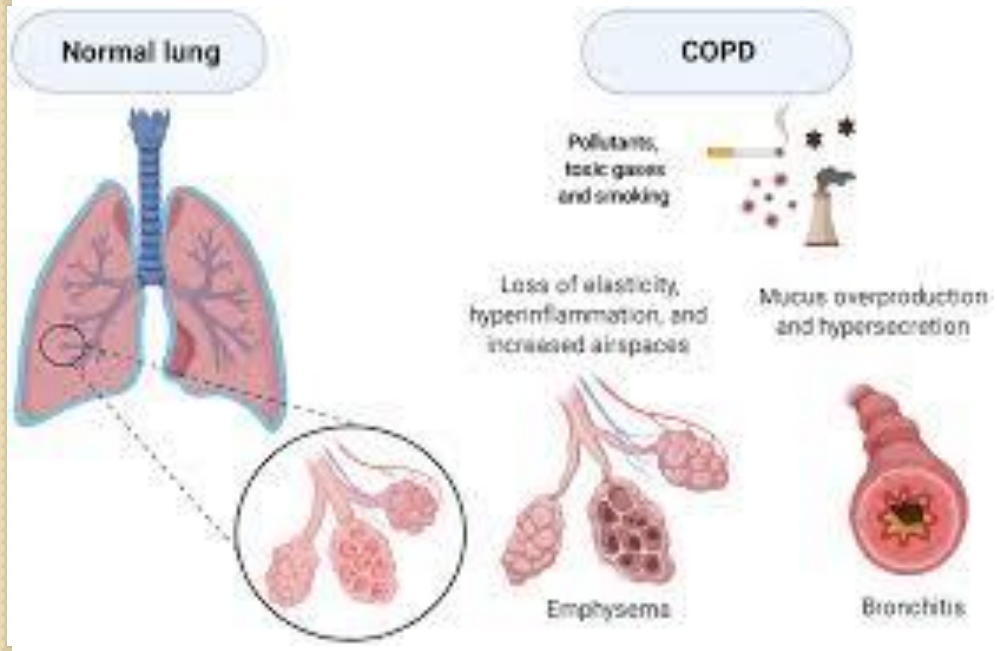
2. **Non-smoking** (10–20% of cases):
   - Air pollution (outdoor and indoor).
   - Recurrent pulmonary infections.
   - Genetic predisposition as alpha 1-antitrypsin deficiency.

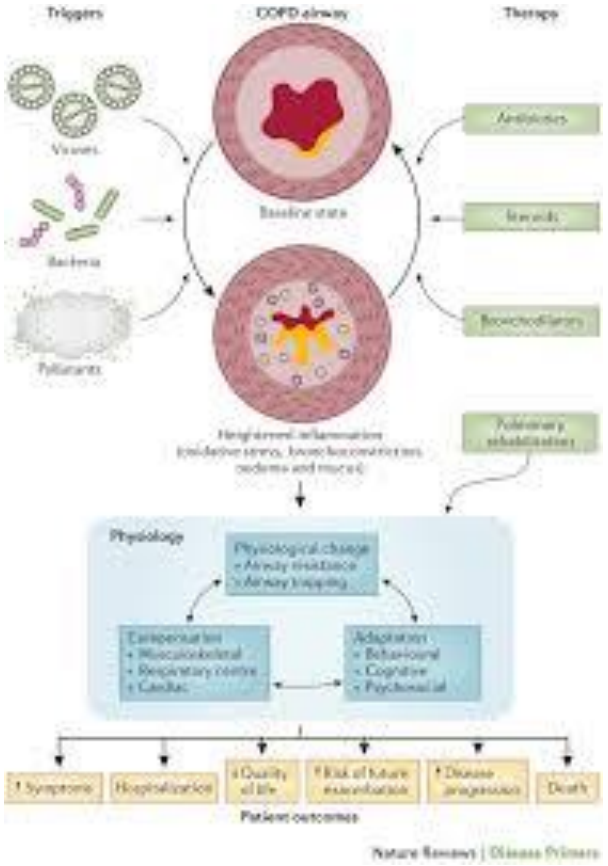
- It is an umbrella term to describe:
  1. Chronic bronchitis, a clinical diagnosis (cough, and sputum >3 months in 2 successive years) which may progress to dyspnea.
  2. Emphysema, an anatomic/ histologic diagnosis, denoting permanent dilatation of pulmonary air spaces distal to the terminal bronchioles.
  3. Asthma-COPD Overlap (ACO); the 2 diseases together.


Emphysema
COPD
Chronic
bronchitis
chronic
Asthma

Pathogenesis and pathology

- After long-term exposure to risk factors, inflammatory cells are recruited in the lungs resulting in chronic airway inflammation.
- This leads to:
  1. Structural airway changes: as proliferation and hyper-secretion of submucosal mucus-secreting glands.
  2. Small airways narrowing:
  3. Destruction of lung parenchyma: inflammatory cells secrete proteinases, which lead to destruction of the acinus (lung structures distal to the terminal bronchiole). Therefore patients with alpha-1 antitrypsin deficiency are more vulnerable to develop emphysema.


Normal lung
COPD
Pollutants, toxic gases and smoking
Loss of elasticity, hyperinflammation, and increased airspaces
Mucus overproduction and hypersecretion
Emphysema
Bronchitis


Triggers
COPD airway
Therapy
Viruses
Bacteria
Pollutants
Baseline state
Antibiotics
Steroids
Bronchodilators
Pulmonary rehabilitation
Physiology
Physiological change
Compensation
Musculoskeletal
Respiratory centre
Cardiac
Adaptation
Behavioural
Cognitive
Psychosocial
↑ Symptoms
Hospitalization
↓ Quality of life
↑ Risk of future exacerbation
↑ Disease progression
Death
Patient outcomes
Nature Reviews | Disease Primers

- The patient may present due to the chronic respiratory symptoms, or due to an acute COPD exacerbation.

Diagnosis of COPD

- The diagnosis of COPD should be considered in anyone with typical symptoms of COPD (cough, sputum production, dyspnea) and/or a history of exposure to risk factors for COPD.

Symptoms and signs of COPD



Barrel Chest





Symptoms of COPD

- Symptoms are minimal or nonspecific until the disease reaches an advanced stage.
- Symptoms of COPD include cough, sputum production, and dyspnea on exertion (all common).
  - Cough usually precede the dyspnea by many years.
  - Expectoration typically occurs in the morning. It is usually scanty white and mucoid. It increases in amount and turns yellow muco-purulent in bacterial exacerbations.
  - Dyspnea is exertional and progressive over years.

Clinical picture (symptoms and signs)

- **Symptoms**

  * **Dyspnea:** is the most prominent symptom in patients with type A

  * **Cough:** is the most common symptom in patients with COPD.

  * **Sputum production.**

  Type B patients generally complain of chronic cough and sputum production.

some patients with COPD are **symptom free**, and the diagnosis is determined on the basis of **pulmonary function tests.**

- The diagnosis should be confirmed by spirometry,
- Showing: post-bronchodilator FEV1/FVC <0.70 (ratio of forced expiratory volume in the first second FEV1 to forced vital capacity FVC < 70%).



- **different Stages of severity in COPD**

  *Post-bronchodilator FEV1/FVC < 70% with

  - **Stage 1. Mild** FEV1 >80% of predicted
  - **Stage 2. Moderate** FEV1 50% to < 80% of predicted
  - **Stage 3. Severe** FEV1 30% to 49% of predicted
  - **Stage 4. Very severe** FEV1<30% of predicted

C

		PRE-DRUG PREDICTED	ACTUAL	% PREDICTED
		Ref	Pre	% Ref
Spirometry				
FVC	Liters	4.35	4.84	111
FEV1	Liters	3.01	2.36	78
FEV1/FVC	%	70	49	
FEF25-75%	L/sec	2.83	0.88	31

Classification of COPD

- According to Global Initiative for Chronic Obstructive Lung Disease (GOLD):
  - **GOLD spirometric grades:** 1,2,3,4 according to post-bronchodilator FEV1% of predicted value. It is used to indicate prognosis.
  - **GOLD groups:** A,B, E according to severity of symptoms and history of exacerbations. It is used to guide treatment.

- According to clinical picture:

pink puffer or blue bloater.

Pink puffer (type A)





Blue bloater (type B)

Concomitant chronic diseases (comorbidities) of COPD

- Patients with COPD often have concomitant chronic illnesses at the time of diagnosis. This may be due to common risk factors (e.g. aging smoking, inactivity). COPD itself also has extra-pulmonary systemic effects.
- Interaction between COPD drugs and other drugs must be considered.

These co-morbidities include:

1. Anxiety and depression
2. Cardiovascular disease (ischemia, arrhythmia, Hypertension)
3. Osteoporosis
4. Malnourishment
5. Skeletal muscle dysfunction
6. Bronchiectasis
7. Obstructive sleep apnea
8. Lung cancer

Complications of COPD

1. Acute exacerbation of COPD.
2. Chronic respiratory failure. It is type 2 (hypoxemia with hypercapnia).
3. Right side heart hypertrophy then failure (cor pulmonale).
4. Secondary spontaneous pneumothorax due to rupture of bullae.

Differential diagnosis of COPD

1. Asthma
2. Congestive heart failure
3. Bronchiectasis
4. Tuberculosis
5. Other causes of chronic cough as gastro-esophageal reflux disease (GERD).

Investigations of COPD

- Only spirometry is essential for the diagnosis of COPD.
- Other tests are to exclude other diagnosis, assess co-morbidities and complications.

1. Pulmonary function tests
2. X-ray chest
3. Blood gases (pulse oximetry and arterial blood gases)
4. Complete blood picture CBC
5. Serum alpha-1 antitrypsin level measurement
6. CT chest
7. ECG and echocardiogram
8. Investigations specific to co-morbidities



Treatment of COPD

- **Treatment of COPD includes 4 items:**
  1. Education about COPD and smoking cessation
  2. Treatment of acute exacerbations
  3. Chronic management of COPD (stable)
  4. Interventions that decrease mortality and increase survival in COPD

1- Education about COPD

This includes education about:

- Avoidance of risk factors,
- Importance and methods of smoking cessation,
- Treatment options,
- Importance and correct use of inhalers,
- Action plan for exacerbations,
- prognosis.

Causes and symptom of acute exacerbation

- Causes: viral or bacterial respiratory tract infections are the most common, followed by bouts of increased air pollution.
- Symptoms: increased **dyspnea** is the key symptom of an exacerbation. Others include increased **sputum** volume or purulence, increased **cough** and **wheezing**.

2- Treatment of acute exacerbations

- Systemic steroids as oral prednisolone (20-40 mg for 5-10 days).
- Inhaled short-acting bronchodilator: beta agonist (SABA) as salbutamol, or short-acting anticholinergic as ipratropium. Also, they can be combined.

  Metered dose inhaler (MDI) if good hand-breath coordination, MDI with spacer if bad coordination, or nebuliser.
- Antibiotics if bacterial infection is suspected.

- Supplemental oxygen when there is hypoxia, titrated to target 90–92% SpO2 not more.
- Noninvasive ventilation (NIV).
- Consider invasive mechanical ventilation if NIV failed or is contraindicated, and the patient's condition and circumstances justify using the invasive route.







3- Management of stable COPD

- The primary goal is to achieve <u>smoking cessation</u> and the use of bronchodilators
- **1- Pharmacological Agents Bronchodilators, Inhaled corticosteroids (ICS), Oxygen therapy**
- **2- Pulmonary Rehabilitation Programs**
- **3- Surgery**



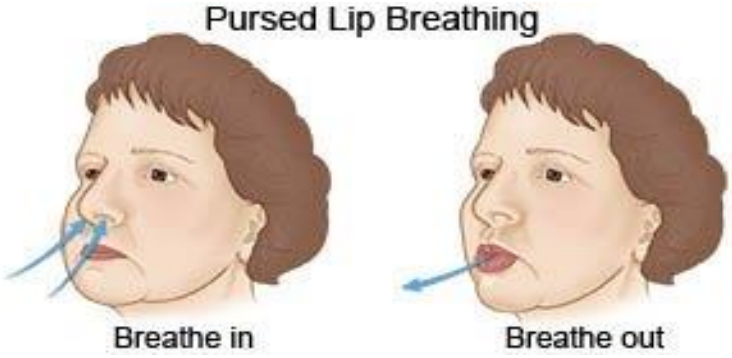






Management of stable COPD

1. Pulmonary rehabilitation:
   - It improves symptoms, exercise capacity, and quality of life.
   - Encourage physical activity.
   - An easy exercise is pursed lip breathing; the patient breathes in through the nose and breathes out slowly through pursed lips. This style of breathing increases airway pressure and prevents bronchial collapse during the last phase of expiration.



Pulmonary Rehabilitation Programs.

It improves exercise tolerance and educates patients about their disease and how to manage it.

Although lung function does not improve with pulmonary rehabilitation, the over all quality of life appears to improve, as evidenced by less dyspnea and exercise tolerance.

2. bronchodilator drugs

1- Sympathomimetics

2- Anticholinergics

3-Theophylline

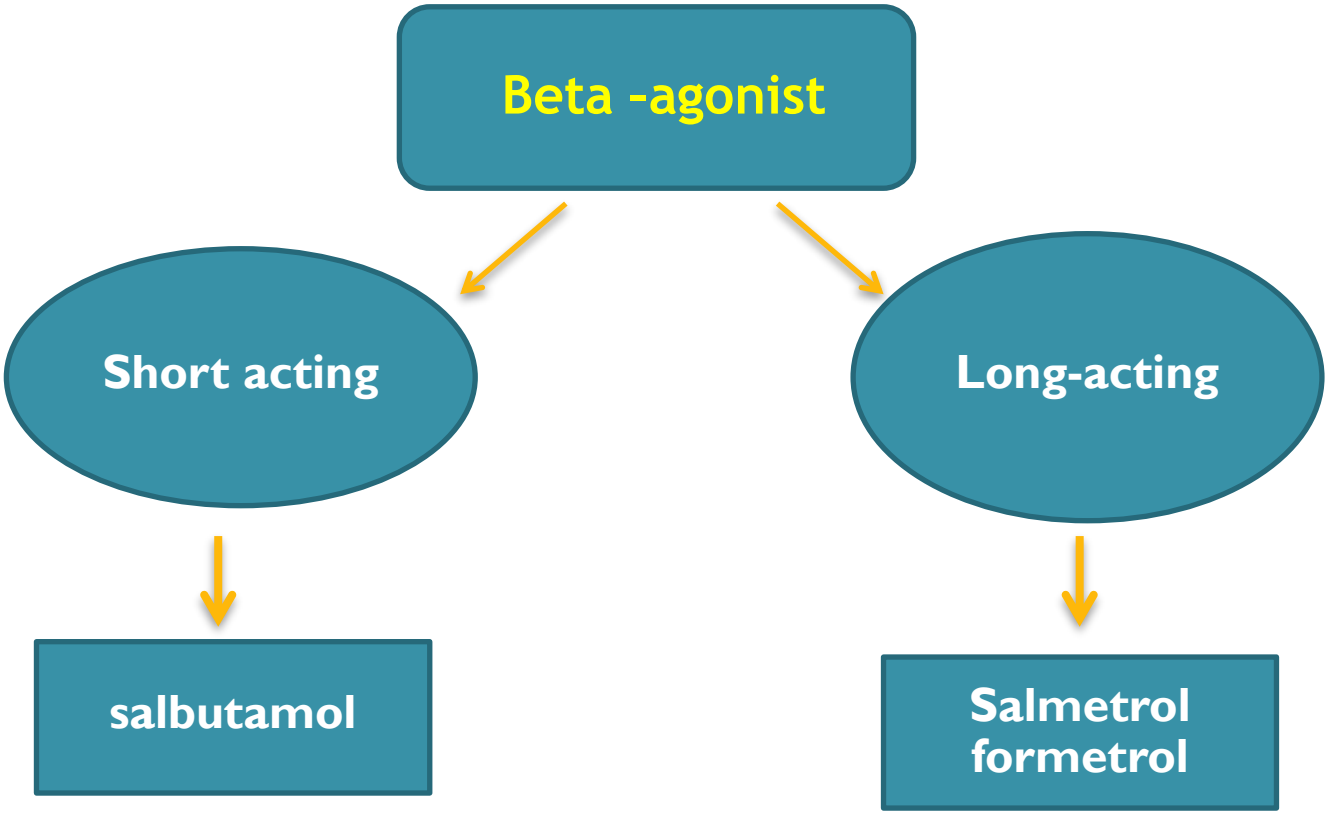
Pharmacological Agents

- COPD patient with moderate to severe disease (stages 2 and 3) usually controlled by pharmacological agents:

1) **Bronchodilators**: do not change the long –term deterioration in lung function.

   They play a key role in reducing symptoms such as dyspnea and improving the quality of life for COPD patients.

- The initial medication used is either:
  a **Beta –agonist** or **anticholnergic**.



Available bronchodilator drugs

1- Sympathomimetics:

- Selective B2 agonists: act on β2 receptors of the bronchi.

- Available as:

A - Oral:

- Some are short acting (6h) as salbutamol.
- Some are long acting as Bambuterol

B - Inhalation:

- Some are short acting (6h) as salbutamol.
- Some are long acting (12h) Long Acting Beta Agonists (**LABA**) as formoterol and salmeterol.
- Some are ultra-long acting beta agonists (24h) as Indacaterol and Vilanterol
- Inhalation therapy must be encouraged as it works rapidly within 5-10 minutes and avoids side affects of systemic use.
- Available as :-

  - Metered dose inhaler: but needs hand-breath coordination

  - Dry powder inhalers: but needs sufficient inspiratory flow

  - Nebulizer

Inhalation therapy must be encouraged as it works rapidly within 5-10 minutes and avoids side affects of systemic use.





A sample of some of the many dry powder inhalers available in the United States



Adverse effects

- Stimulation of beta2-adrenergic receptors can produce resting sinus tachycardia and has the potential to precipitate cardiac rhythm disturbances in susceptible patients.
- Exaggerated somatic tremor is troublesome in some older patients.

2- Anticholinergics:

- Have atropine-like action
- Available as inhalation.
- Short acting Ipartropium bromide
- Long acting Tiotropium, Glycopyrronium, Umeclidinium

Anticholinergic

Short acting

ipratropium

Long-acting

tiotropium



Adverse effects

- Anticholinergic drugs are poorly absorbed which limits the troublesome systemic effects seen with atropine.
- The main side effect is dryness of the mouth.
- occasional prostatic symptoms have been reported
- Use of solutions with a face mask has been reported to precipitate acute glaucoma

3- Theophyllin:-

- Act directly on smooth muscles of bronchi.
- Available as:
  - Injection
  - Oral: some are long acting sustained release
  - Suppository

- Methylxanthines: such as theophylline need carful monitoring due to narrow therapeutic index.

Side effects of Methylxanthines

- Methylxanthines are nonspecific inhibitors of all phosphodiesterase enzyme subsets, which explains their wide range of toxic effects.
- Problems include the development of atrial and ventricular arrhythmias (which can prove fatal) and grand mal convulsions .
- Other side effects include headaches, insomnia, nausea, and heartburn, and these may occur within the therapeutic range of serum theophylline.

Inhaled corticosteroids(ICS);

Inhaled route is preferred - Avoid systemic use except in exacerbations.

Available as:-

- Oral: Prednisolone for exacerbations only
- Inhalation (inhaled cortico-steroids) (**ICS**):

  - effective and avoids side effects of systemic steroids

  - As: Beclomethasone, Budesonide, Fluticasone, ciclesonide

- indicated in severe and very severe grades with frequant exacerbationas as it proved to decrease the frequancy of acute exacerbationas

Adverse effects of Inhaled corticosteroid

- is associated with higher prevalence of oral candidiasis, hoarse voice, and skin bruising.
- Treatment with inhaled corticosteroids is associated with an increased risk of pneumonia.
- While long-term treatment is associated with an increased risk of reduced bone density.

How to Manage stable COPD

1. Short acting beta agonists (SABA), short acting muscarinic antagonists (SAMA), or both to relieve intermittent increases in dyspnea.

2. If symptoms persist, give long acting beta agonists (LABA), or long acting muscarinic antagonists (LAMA), or both (combined bronchodilator LABA/LAMA).

3. If symptoms still present or exacerbations are frequent in spite of using LABA/LAMA, then add inhaled corticosteroids (ICS). This is called triple therapy (LABA/LAMA/ICS).

4. Give combined ICS with a LABA (combined ICS/LABA) since the start if there are signs of eosinophilic inflammation or asthmatic component to the COPD. But notice side effects of ICS, specially at higher doses: dysphonia, oral candidiasis, cataract, skin bruising, increased pneumonia, and diminished bone density.

5. Vaccination: influenza vaccine/year, pneumococcal vaccine, COVID-19 vaccine.

6. Drugs for special situations:

- Phosphodiesterase-4 inhibitor (Roflumilast) in chronic bronchitis COPD phenotype with frequent exacerbations, to decrease future exacerbations.
- Azithromycin long term in COPD patients with frequent exacerbations to decrease exacerbations, but it is associated with increased cardiovascular risks.
- Theophylline may be tried if other bronchodilators are not available; the benefit is unproven.
- Mucolytics liquefy mucus, thus making it easier to cough up. This may reduce exacerbations of COPD in some patients.
- α1-antitrypsin augmentation therapy for patients with AATD.

4- Interventions that decrease mortality and increase survival in COPD

1. Home oxygen
2. Smoking cessation
3. Lung volume reduction maneuvers
4. Lung transplantation

- The goal of oxygen therapy is to keep PaO2 above 60 mmHg and /or the SaO2 above 90%.

- oxygen therapy improve survival, exercise tolerance, and hemodynamic in COPD Pts.

- Short –term oxygen therapy may be useful for treatment of the intermittent desaturations seen during exercise or sleep in selected cases.

Surgery

It my prove useful in treating the patient with severe COPD.

Lung volume reduction surgery (LVRS); remove the most diseased parts of the lung that are restricting function of the lung.

bollectomy; excision the huge emphysematous bulla.

Lung transplantation.


THANK YOU!